

Static water-table baseline corrected by observed regional storage change

Static WTD baseline
Fan et al. layer

Regional storage trend
GRACE/GRACE-FO ~300 km

Delta WTD
Delta TWS / S_y

Water-table term only
Zhu et al. screen

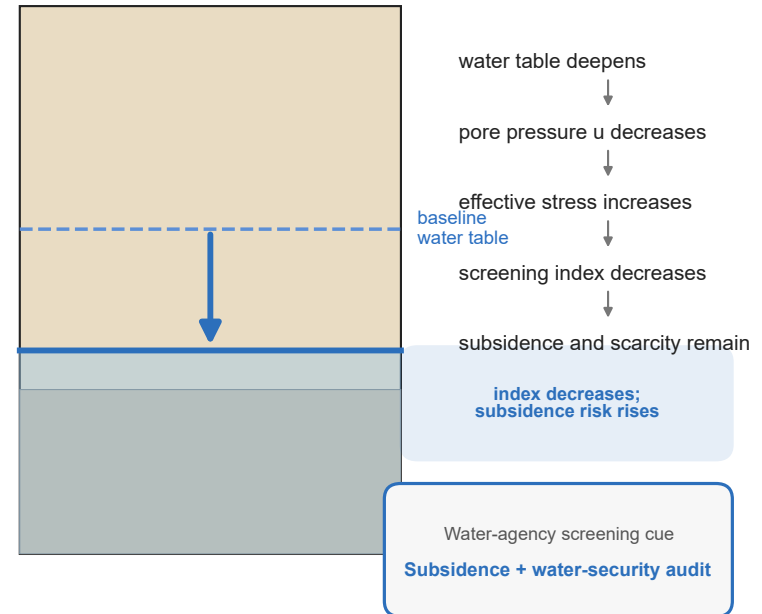
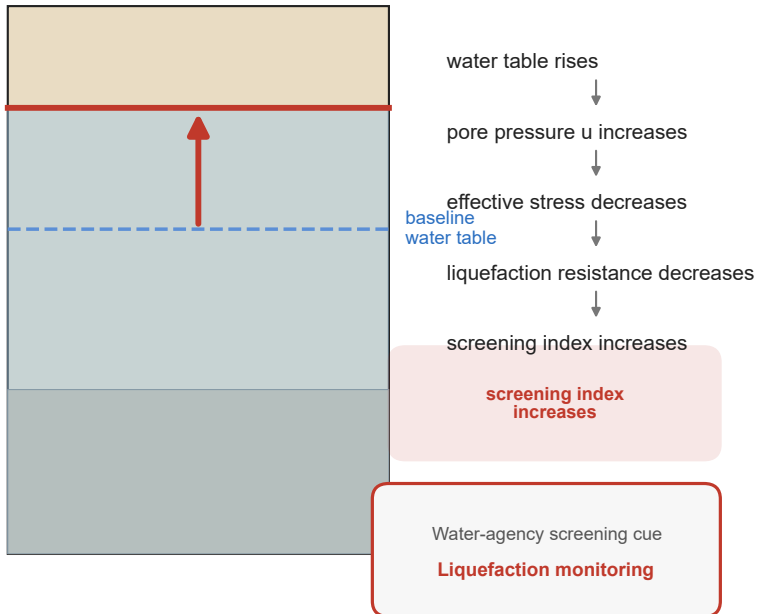
Follow-up flag
local wells + sediments

Recharge or aquifer recovery

Depletion or over-abstraction

North China Plain / Beijing

Punjab / Delhi-Lahore-Ludhiana



The mechanism changes liquefaction susceptibility if shaking occurs; it does not imply that groundwater causes earthquakes.